



REPORT OF INDIVIDUAL DESIGN PROJECT

INTAKE 41

GASGUARD – IoT and ML Based Gas Cylinder Monitoring & Prediction with Automatic Servo Controlled Shutoff

BY
WALP Harsha
D/ENG./24/0095/ET

Department of Electrical, Electronic and Telecommunication Engineering

General Sir John Kotelawala Defence University Sri Lanka

June 2026

ABSTRACT

Urban households and commercial users of liquefied petroleum gas (LPG) face significant challenges in ensuring safe usage and accurate monitoring of cylinder levels due to the lack of continuous supervision and predictive awareness. Manual monitoring techniques are frequently unreliable, leading to unexpected gas depletion, safety hazards such as gas leakage, fire risks, and inefficient energy management. Existing systems are generally limited to basic alarm functions and do not provide intelligent monitoring, remote access or automated safety control, making them unsuitable for modern smart environments.

This project “**GasGuard : IoT-ML Based Smart LPG Monitoring and Safety System**” proposes an automated and intelligent solution for real-time gas leakage detection, cylinder weight monitoring, and predictive gas usage analysis. The system is built using an ESP32 Dev kit V1 micro controller integrated with a load cell and HX711 module to continuously measure LP Gas cylinder weight and an MQ-5 gas sensor to detect hazardous gas leakage conditions. A MG995 servo motor is used to automatically shut off the gas supply during emergency situations, while a buzzer, LED s and LCD Display provide immediate local alerts.

The system is powered through a 230V AC mains supply with a regulated power supply unit using buck converter to ensure stable and continuous operation. Sensor data is transmitted to Google Firebase for real-time cloud storage and remote monitoring. A cross-platform mobile application developed using React Native with Kotlin and Java native modules provides live dashboards, alerts, usage graphs, cost estimation and predicted refill dates and enables users to place LPG cylinder refill orders directly through the application by providing their delivery location, ensuring timely replacement and uninterrupted gas availability. A machine learning-based model analyzes historical consumption data to predict gas depletion trends and estimate remaining cylinder life for proactive refill planning.

The proposed system improves safety, reduces manual intervention, enhances energy efficiency and enables intelligent predictive monitoring for modern LPG management systems. The complete prototype was developed with an estimated hardware implementation cost of **Rs. 8,290** making it a low-cost and practical solution for domestic and small commercial LPG users.

Keywords: LPG Monitoring, ESP32, MQ-5 Sensor, Load Cell, HX711, Firebase, React Native, Mobile Application, Predictive analytics, Automatic Gas Shut-off, LPG Cylinder Ordering, IoT, Gas Leakage Detection, Machine Learning, Smart Safety System

TABLE OF CONTENTS

ABSTRACT.....	2
TABLE OF CONTENTS	3
LIST OF TABLES	5
LIST OF FIGURES.....	6
CHAPTER 1: INTRODUCTION	7
1.1 BACKGROUND	7
1.2 PROBLEM STATEMENT	7
1.3 SCOPE OF THE PROJECT	7
CHAPTER 2: LITERATURE REVIEW.....	8-12
2.1 INTRODUCTION.....	8
2.2 REVIEW OF RELATED STUDIES.....	8
2.3 HOW THE PROPOSED SYSTEM DIFFERS FROM PREVIOUS WORK.....	9
2.4 SOURCES RELEVANT TO DESIGN CHOICES.....	10
2.5 MARKET SURVEY	11
2.6 OVERVIEW OF MARKERT TRENDS.....	11
2.7 EXISTING TECHNOLOGIES AND THEIR LIMITATIONS	11
2.8 POTENTIAL USER BASE AND APPLICATION AREAS.....	12
2.9 SUMMARY	12
CHAPTER 3: OBJECTIVES AND AIMS	13
3.1 OBJECTIVES	13
3.2 AIM	13
CHAPTER 4: DESIGN AND METHODOLOGY.....	14-29
4.1 INTRODUCTION.....	14
4.2 SYSTEM ARCHITECTURE.....	14
4.3 COMPONENTS OF GASGUARD SYSTEM	14
4.4 BLOCK DIAGRAM OF THE PROPOSED SYSTEM.....	17
4.5 FLOWCHART OF SYSTEM OPERATION	18
4.6 FLOWCHART OF MOBILE APPLICATION	18
4.7 CALIBRATIONS	19
4.7.1 Load Cell Calibrations	19
4.7.2 MQ - 5 Sensor Calibrations	19
4.8 CIRCUIT DESIGN	20
4.9 GASGUARD MONITORING NODE DESIGN.....	20
4.10 GAS GUARD CONTROL UNIT DESIGN	21
4.11 NETWORK COMMUNICATION PROTOCOLS ,CLOUD ARCHITECTURE & LOCATION SHARING	21
4.12 COMMUNICATION SYSTEM DESIGN	22
4.13 SOFTWARE IMPLEMANETATION	22
4.13.1 Embedded System Programming	22
4.13.2 Firebase Integration	22
4.13.3 Mobile Application Development.....	23

4.14 CRITICAL DESIGN DECISION	23
4.15 CUSTOM WOODEN ENCLOSURE DESIGN	23
4.16 PROTOTYPE DEVELOPMENT	24
4.17 FINAL IMPLEMENTATION	25
4.18 MACHINE LEARNING BASED DEPLETION PREDICTION.....	25
4.18.1 Alogorithm Selection and Theoretical Basis	25
4.18.2 Data Collection and Feature Eengineering Pipeline.....	25
4.18.3 Model Training and Deployment Setup.....	26
4.19 COST ANALYSIS	26
4.20 COST-SAVING MEASURES IMPLEMENTED	27
4.21 ASSUMPTIONS AND LIMITATIONS	27
4.21.1 Machine Learining Performance	28
4.21.2 System Power Consumption.....	28
4.21.3 End - to -End Alert Latency System	29
4.21.4 Overall System Performance	29
CHAPTER 5: FINAL OUTCOME AND DISCUSSION.....	30-33
5.1 INTRODUCTION.....	30
5.2 FINAL OUTCOME.....	30
5.3 SYSTEM OPERATION RESULTS.....	31
5.3.1 Model Training Results.....	31
5.3.2 Firebase Integration Results.....	31
5.3.3 Mobile Application Results	32
5.3.3.1 Customer Side Mobile Application	32
5.3.3.2 Dealer Side Mobile Application	33
5.4 DISCUSSION.....	34-38
5.4.1 RESULTS	34
5.4.2 DEPLETION PREDICTION MODEL RESULTS AND ANALYSIS	34
5.4.2.1 Training Convergence and Loss Analysis.....	34
5.4.2.2 Evalutaion Metrics and Cross-Validation	35
5.4.2.3 Model Comparison and Feature Importance	36
5.5 PROBLEMS ENCOUNTERED AND SOLUTIONS	37
5.6 TEST RESULTS SUMMAR.....	38
CHAPTER 6: CONCLUSIONS AND FUTURE WORKS.....	39-40
6.1 CONCLUSION	39
6.2 PROJECT ACHIEVEMENTS	39
6.3 FUTURE WORKS	40
6.4 FINAL REMARKS	40
ANNEXURES	41-43
ANNEXURES A: SOURCE CODE REPOSITORY	41
ANNEXURES B: GAS GUARD MACHINE LEARNING MODEL STRUCTURE	42
ANNEXURES C: GAS GUARD MOBILE APP STRUCTURE	42
ANNEXURES D: LIBRARY DEPENDENCIES	43

ANNEXURES E: COMPONENT DATASHEETS	43
REFERENCES	44

LIST OF TABLES

Table 2.1: Summary of Related Literature.....	9
Table 4.1:MQ-5 Gas Leak Detection Calibration Results	19
Table 4.2: Critical Design Decisions	23
Table 4.3: Component List and Cost Estimation	27
Table 4.4: Machine Learning Model Performance Summary	28
Table 4.5: System Power Consumption Summary.....	28
Table 4.6: End-to-End Alert Latency Analysis	29
Table 4.7: Overall System Performance Summary	29
Table 5.1: Functional Validation Summary of The Gas Guard Mobile APP	33
Table 5.2: Model Comparison	36
Table 5.3: Summary of Test Result	38
Table 7.1: Library Dependencies	43
Table 7.2:Component Datasheet	43

LIST OF FIGURES

Figure 2.1: Importance of Conserving Gas in Daily Usage.....	10
Figure 2.2: Usefulness of a Gas Level Display Device	10
Figure 2.3: Likelihood of Using an Easy , affordable Product.....	10
Figure 2.4: Preferred Method for Displaying Gas Levels	10
Figure 2.5: Gas Cylinder market size, 2023 – 2033 (USD Billion).....	11
Figure 2.6: Existing Products	12
Figure 4.1: System Block Diagram.....	17
Figure 4.2: Flowchart of the GasGuard System Operation	18
Figure 4.3: Flowchart of the GasGuard Mobile Application	18
Figure 4.4: Load Cell Calibration Graph.....	19
Figure 4.5: Schematic Diagram of PCB	20
Figure 4.6: 3D View of PCB Layout View.....	20
Figure 4.7: PCB Figure	20
Figure 4.8: 3D View of PCB.....	20
Figure 4.9: System Design of the GASGUARD Embedded Monitoring Node.....	20
Figure 4.10: System Design of Control Unit.....	21
Figure 4.11:Communication System Design of the GASGUARD	22
Figure 4.12: Custom Wooden Enclosure Design	24
Figure 4.13: Prototype Assembly Process.....	24
Figure 4.14: Final Product Design.....	25
Figure 5.2: Firebase Integration Results.....	31
Figure 5.3:Customer Side Mobile App Dashboard	32
Figure 5.4: Dealer Side Mobile App Dashboard.....	33
Figure 5.5: Training Convergence Log	34
Figure 5.6:Evaluation & Performance Metrics.....	35
Figure 5.7:Model Comparison & Final Equation.....	36
Figure Annexure A: Source Code Repository (Full)	41
Figure Annexure B: GasGuard Machine Learning Model Strucuture.....	42
Figure Annexure C:GasGuard Mobile App Strucuture.....	42

CHAPTER 1: INTRODUCTION

1.1 BACKGROUND

Liquefied petroleum gas (LPG) is one of the most widely used energy sources for household and commercial cooking due to its efficiency, convenience, and affordability. In developing regions such as Sri Lanka, many households depend on LPG for daily activities. However, increasing usage and busy lifestyles have reduced user awareness of cylinder levels and safety conditions.

In practical situations, users often realize gas depletion only when the cylinder becomes empty during cooking, causing inconvenience. Additionally, LPG leaks caused by faulty regulators, valves, or hoses can lead to serious fire and explosion risks. Reports from the Engineering Institution of Sri Lanka indicate that more than 400 gas-related incidents were recorded within a short period in 2021 [1]. This ongoing safety concern was further highlighted in June 2026 when a devastating fire at the “Mawpiya Sevana” Elderly Care Home in Anguruwatota, Horana resulted in the loss of 13 residents [3], emphasizing the importance of early warning and monitoring systems [2].

LPG mainly consists of highly flammable gases such as propane and butane, which require proper handling and continuous monitoring. Since conventional LPG systems lack automatic safety features, there is a need for an affordable intelligent solution that can detect leaks, monitor cylinder levels, and provide real-time safety alerts to users.

1.2 PROBLEM STATEMENT

The core problem addressed by this project can be summarised as follows:

- Existing LPG monitoring methods are manual and unreliable, causing unexpected gas depletion due to the lack of real-time level tracking.
- Gas leaks often remain undetected without continuous monitoring, increasing the risk of fires, explosions, and health hazards.
- Gas leaks often remain undetected without continuous monitoring, increasing the risk of fires, explosions, and health hazards.
- An affordable integrated system is required to combine LPG level monitoring, leak detection, predictive analysis, automatic safety control, and remote notifications in a single unit.

1.3 SCOPE OF THE PROJECT

This project covers the design, development, and testing of an IoT & ML - based smart LPG monitoring and safety system. The scope includes real-time cylinder weight measurement using a load cell, gas leak detection, cloud-based data monitoring, mobile notifications, LPG consumption prediction, and automated valve shutoff for improved user safety and management.

CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION

Liquefied Petroleum Gas (LPG) is widely used in residential and commercial applications due to its efficiency and affordability. However, LPG leaks can cause serious hazards such as fires, explosions and health risks, requiring reliable monitoring solutions [1],[2]. Advances in IoT, embedded systems, and mobile technologies have enabled smart gas monitoring systems with real-time leak detection, gas level monitoring, cloud connectivity and mobile alerts [3],[4]. This chapter reviews existing LPG leak detection and monitoring systems, highlighting their limitations and the need for an integrated approach [5],[6]. The proposed **GasGuard** system combines gas leak detection, cylinder weight monitoring, automatic valve control, Firebase based cloud monitoring, and a mobile application for real-time safety management.

2.2 REVIEW OF RELATED WORK

Several prior works have explored accident detection and alert systems. The following table summarises six relevant studies, the technologies they used, and their key limitations.

Ref	Title & Year	Technologies	Key Limitations
[1]	Automatic Gas Leakage Monitoring System Using MQ-5 Sensor (2021)	MQ-5 Gas Sensor, Arduino UNO, GSM Module, LCD, Buzzer	Detects gas leaks only; no cylinder level monitoring, predictive analytics, or automatic valve shutoff; depends on GSM network for notifications.
[2]	IoT Based LPG Detection and Interfacing with ThingSpeak – IRJET (2020)	MQ-5 Gas Sensor, Arduino UNO, ESP8266 Wi-Fi Module, ThingSpeak Cloud, SMS Alert System	No automatic gas shutoff; lacks load-cell-based cylinder weight monitoring; cannot estimate remaining LPG or consumption.
[3]	IoT-Based System for Detecting and Monitoring LPG Leaks in Residential Settings (2025)	MQ-5 Sensor, ESP32, MQTT Protocol, Cloud Platform, Wi-Fi, Relay Module, Buzzer, LEDs	Does not measure cylinder level or predict LPG usage; sensor calibration challenges; cannot estimate remaining gas quantity.
[4]	IoT-Based Gas Leakage Detection and Gas Empty Alert System (2025)	MQ-135 Gas Sensor, Arduino, Load Cell, GSM Module, Buzzer	Dependent on power and internet/GSM connectivity; requires sensor calibration; load-cell accuracy limitations.

[5]	IOT Based Gas Leakage Detection and Gas Empty Alert System (2021)	Arduino MKR WiFi 1010, MQ-6 Sensor, SEN-10245 Load Cell, Wia IoT Cloud, Twilio API, Wi-Fi, Exhaust Fan, Buzzers	Internet and power dependent; notification delays due to network issues; requires sensor calibration; limited load-cell accuracy.
[6]	Estimation and Prediction of LPG Gas Using Data Analytics (2022)	NodeMCU, MQ-6 Gas Sensor, Load Cell, Google Firebase, GSM Module, MIT App, LCD, Data Analytics	Depends on internet and GSM; no automatic gas shutoff; sensor calibration and load-cell accuracy issues; prediction errors and power dependency.

Table 2.1: Summary of Related Literature

A consistent gap identified across these studies is the lack of an integrated, low-cost system that combines **real-time gas leak detection, cylinder weight monitoring, LPG consumption prediction, automatic gas valve shutoff, mobile application support, and cloud-based data logging** within a single IoT-enabled platform. Several existing solutions focus only on leak detection or cylinder level monitoring, while predictive analytics, remote monitoring, and automated safety mechanisms are often absent or implemented separately.

2.3 HOW THE PROPOSED SYSTEM DIFFERS FROM PREVIOUS WORK

The proposed GasGuard system addresses the limitations of existing LPG monitoring solutions through the following key features :

1. Integrated gas leak detection and cylinder weight monitoring using an MQ-5 gas sensor and HX711 load cell in a single embedded system.
2. Automatic emergency valve shutoff using a servo motor to minimize gas leakage immediately after leak detection.
3. Real-time Firebase integration for remote monitoring of gas level, cylinder weight, temperature, humidity, and system status through a mobile application.
4. Local safety alerts using a buzzer, LEDs, and LCD display to provide immediate visual and audible warnings during hazardous conditions.
5. A low-cost IoT-based architecture built on the ESP32 platform, making the system affordable, scalable, and suitable for residential and small commercial LPG safety applications.

2.4 SOURCES RELEVANT TO DESIGN CHOICES

The system design was designed based on IoT based gas monitoring studies, [1] and [3] for LPG leak detection and firebase cloud integration. For accurate estimation of the cylinder weight, load cell-based systems were investigated [4], [5] and embedded safety systems were utilized to develop automatic alert mechanisms and emergency response features [2], [6]. Furthermore, mobile IoT applications [3] and [4] assisted with the real-time monitoring and notification to users.

2.5 MARKET SURVEY

The smart LPG monitoring market is growing due to three key trends. First, there is an increasing demand for real-time gas leak detection and safety systems to prevent domestic accidents. Second, the adoption of IoT and cloud technologies is enabling remote monitoring, predictive gas consumption analysis, and mobile notifications. Third, affordable aftermarket smart LPG monitoring solutions are becoming more popular, as conventional LPG cylinders lack built-in intelligent monitoring and safety features.

Market Survey Results

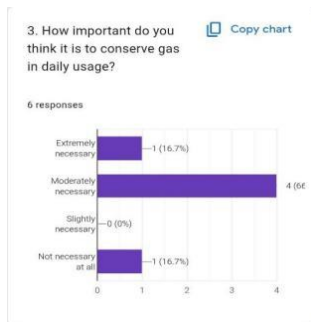


Figure 2.1- Importance of Conserving Gas in Daily Usage

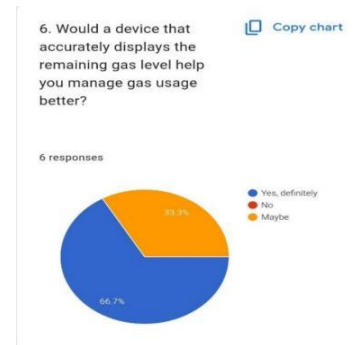


Figure 2.2 - Usefulness of a Gas Level Display Device



Figure 2.3 - Likelihood of Using an Easy , affordable Product

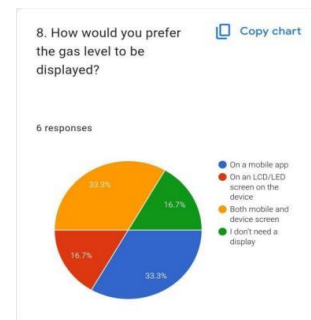


Figure 2.4 - Preferred Method for Displaying Gas Levels

A Google Forms survey of six domestic LPG users showed that all respondents (100%) use LPG cylinders, while 84% considered gas level monitoring necessary. The proposed IoT-based system received positive feedback, with 67% willing to use it and 100% supporting mobile-based cylinder ordering. Users preferred real-time monitoring, mobile alerts, usage tracking, and refill prediction. These findings confirmed the demand for the GasGuard system and guided its final design.

2.6 OVERVIEW OF MARKERT TRENDS

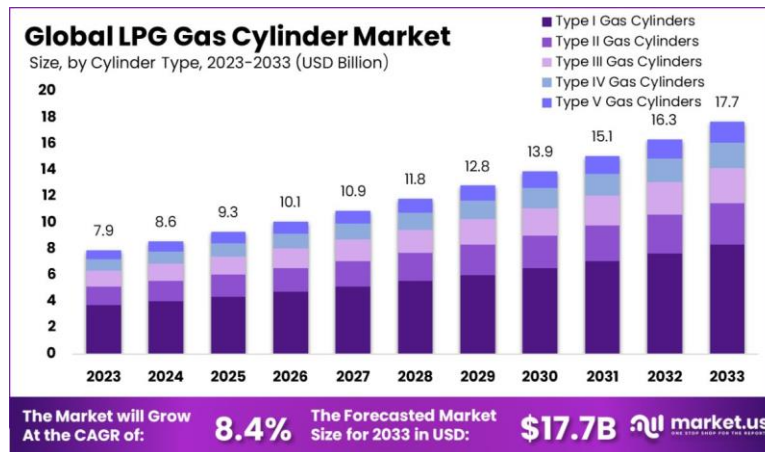


Figure 2.5 -Gas Cylinder market size, 2023 – 2033 (USD Billion)

The consumer utility and safety sector is rapidly adopting smart technologies, particularly for LPG cylinder monitoring and gas leak detection [2]. Increasing demand for improved fuel management, real-time monitoring, and accident prevention has accelerated the adoption of IoT-based gas safety systems[3],[4].The global LPG market, valued at **USD 163.06 billion** continues to grow, highlighting the need for intelligent solutions that replace manual cylinder checking with automated monitoring and safety features [1],[2].

The Asia-Pacific region is the fastest-growing market for LPG cylinders and smart gas monitoring, accounting for **61% of global cylinder demand** and **47% of LPG consumption** [2],[3].Rapid urbanization, increasing household LPG usage, and government clean-energy initiatives in countries such as China and India are driving the adoption of smart monitoring technologies [3].Asia-Pacific also leads the global market with a **34.8% revenue share**, while the wireless IoT gas monitoring segment is projected to grow at a **14.1% CAGR** through 2034 [4],[5].This growth reflects the increasing transition from conventional LPG systems to intelligent, predictive gas safety solutions [2], [5].

2.7 EXISTING PRODUCTS AND THEIR LIMITATIONS

Existing LPG safety products mainly focus on gas leak detection and basic alerts, but lack integrated smart monitoring features.

- **FORENSICS DETECTORS FD-91-RED:** Offers portable combustible gas detection with LCD display and alarms, but lacks IoT connectivity, remote monitoring, LPG level estimation, predictive analysis, and automatic safety control.
- **Cipe LIFE Smart Gas Detector:** Provides Wi-Fi alerts, battery monitoring, and physical alarms. However, it lacks LPG level monitoring, load cell integration, usage prediction, automatic shutoff, and consumption analytics.
- **GasDog GD1000-GS91 WiFi Gas Leak Monitor:** Supports multi-gas detection, mobile alerts, and alarm notifications, but does not provide LPG level tracking, refill prediction, usage analytics, or automatic valve control.



Figure 2.6 - Existing Products

The main limitation across existing products is the absence of an affordable all-in-one solution combining **gas leak detection, LPG level monitoring, prediction, automatic shutoff, and mobile connectivity.**

2.8. POTENTIAL USER BASE AND APPLICATION AREAS

2.8.1. POTENTIAL USER BASE

- **Urban and Rural Households:** Improves family safety by detecting gas leaks and enables efficient LPG usage monitoring.
- **Commercial Kitchens and Restaurants:** Prevents unexpected gas depletion and enhances workplace safety.
- **LPG Distributors and Suppliers:** Supports demand prediction and improves cylinder delivery management.

2.8.2. APPLICATION AREAS

- **Residential Kitchens and Apartments:** Provides real-time LPG monitoring and leak protection for households.
- **Small Food Businesses:** Helps manage gas usage and avoid interruptions during operations
- **Smart Home Systems:** Enables remote LPG monitoring and automated safety control.

2.9 SUMMARY

The reviewed literature shows that existing LPG safety systems are mostly limited to either gas leak detection or cylinder level monitoring , but rarely combine both with predictive analytics, automatic shutoff, and mobile cloud integration. The GasGuard system addresses this gap by providing real-time leak detection, LPG weight monitoring, refill prediction, remote alerts, and automated valve control in a single low-cost IoT platform.

CHAPTER 3: OBJECTIVES AND AIM

3.1 AIM

- To improve household safety and energy efficiency in modern kitchens by designing an automated monitoring, control, and predictive analysis system that prevents gas accidents and tracks LPG consumption levels.

3.2 OBJECTIVES

The following specific objectives were defined to achieve the project aim:

1. **LPG Level Monitoring** : Measure LPG cylinder weight using a load cell and HX711 module to estimate remaining gas levels and consumption patterns.
2. **Gas Leakage Detection** : Detect LPG Leaks using an MQ - 5 Gas Sensor and identify hazardous gas conditions in real time.
3. **Cloud Data Management** : Store and manage historical LPG usage data using Firebase for remote monitoring and analysis.
4. **Predictive Analysis** : Develop a machine learning model to predict gas depletion dates and support timely cylinder replacement.
5. **Mobile Application** : Develop a mobile application to display live sensor data, usage graphs, cost calculations, predicted refill dates, and safety alerts.
6. **Automatic Safety Control** : Implement a servo-based gas shutoff mechanism to automatically isolate the gas supply during leakage conditions.
7. **Smart LPG Supply Management** : Notify nearby LPG shop owners when cylinder levels become low and enable households to place refill orders through the mobile application for efficient delivery management.
8. **Environmental Monitoring**: Monitor ambient temperature and humidity using the DHT22 sensor to support accurate LPG safety analysis.

CHAPTER 4: DESIGN AND METHODOLOGY

4.1 INTRODUCTION

This chapter describes the design and implementation methodology of the proposed GasGuard IoT& ML Based LPG Monitoring and Safety System. The system is designed to provide real-time monitoring of LPG cylinder level, gas leakage detection, predictive analysis, and automatic safety control through the integration of embedded sensors, cloud services, a mobile application, and Google Maps for location-based refill support. This chapter presents the overall system architecture, hardware design, software implementation, communication framework, prototype development, and cost analysis.

4.2 SYSTEM ARCHITECTURE

The proposed system consists of two main layers: the **Embedded Monitoring Node** and the **Cloud + User Interface Layer**. The embedded node is built around the ESP32 Dev kit V1 which is connected to a load cell with HX711 for LPG level measurement, an MQ-5 gas sensor for leak detection, and a servo motor for automatic gas shutoff during emergency conditions. It continuously reads sensor data, detects hazardous conditions, and sends the collected information to the cloud.

The cloud layer receives sensor data, stores historical records in Firebase and supports remote monitoring and analysis through the mobile application. The user interface layer consists of a React Native mobile application that provides a live dashboard, gas usage graphs, refill predictions, safety alerts Google Maps integration for location sharing and LPG delivery support.

4.3 COMPONENTS OF GASGUARD SYSTEM

- **ESP – 32 Dev kit V1 Micro Controller**

The ESP32 DevKit V1 acts as the main controller of GasGuard. It processes sensor readings, controls the LCD, buzzer, LEDs, and MG995 servo motor, and performs real-time safety decisions. Its built-in Wi-Fi also enables sensor data and system status to be synchronized with Firebase.



ESP –32 Dev Kit V1

- **MQ-5 Gas Sensor**

The MQ-5 sensor is used for continuous LPG leakage detection. Its analog output is connected to **GPIO 34** of the ESP32. When the gas reading exceeds the predefined threshold of **500**, the system activates safety alerts and initiates automatic valve closure.



MQ-5 Gas Sensor

- **DHT22 Temperature and Humidity Sensor**

The DHT22 sensor measures temperature and humidity and sends the data to the ESP32 through a digital pin (GPIO27). The ESP32 reads this data using a DHT library.



DHT22 Sensor

- **HX711 Load Cell Amplifier with Load Cell Sensor**

The load cell platform continuously measures the LPG cylinder weight. Changes in weight produce small resistance variations in the internal strain gauges, generating a differential voltage. The HX711 amplifies this signal and converts it into a precise 24-bit digital value. This data is transmitted to the ESP32 through GPIO 25 (DT) and GPIO 26 (SCK), enabling real-time estimation of the remaining LPG quantity and gas consumption.



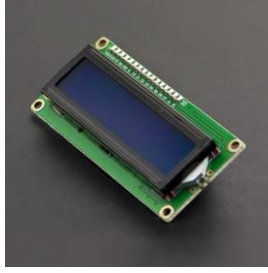
HX711 Load Cell Amplifier



Load Cell 20 Kg

- **LCD Display 16x2**

The 16×2 LCD provides local real-time information such as gas level, cylinder weight, temperature, humidity, and leakage warnings. The ESP32 communicates with the display through the I²C protocol using **GPIO 21 (SDA)** and **GPIO 22 (SCL)** at address **0x27**. Functions such as `lcd.setCursor()` and `lcd.print()` are used to display and update system information on the screen.



LCD Display 16x2 Screen

- **LM2596 DC-DC buck converter step-down module**

The LM2596 module steps down the 12V DC power supply to a suitable voltage level for the ESP-32 dev kit and other components, ensuring safe and stable operation.



LM2596 DC-DC buck converter step-down module

- **MG995 Servo Motor**

The MG995 servo motor controls the gas shut-off valve. When a leak is detected, the ESP32 sends a PWM signal to rotate the servo and close the valve, isolating the gas supply during emergencies.



MG995 Servo Motor

- **LEDs and Buzzer**

The LEDs and buzzer provide local visual and audible alerts. Controlled by the ESP32, they indicate normal and hazardous conditions based on sensor readings and safety thresholds.



Active Buzzer



LEDs

- **12V DC power adapter**

The 12V DC adapter supplies the system, while a step-down converter regulates the voltage to 5V for safe operation of the ESP32 and other components.



12 V DC Power Adapter

4.4 BLOCK DIAGRAM OF THE PROPOSED SYSTEM

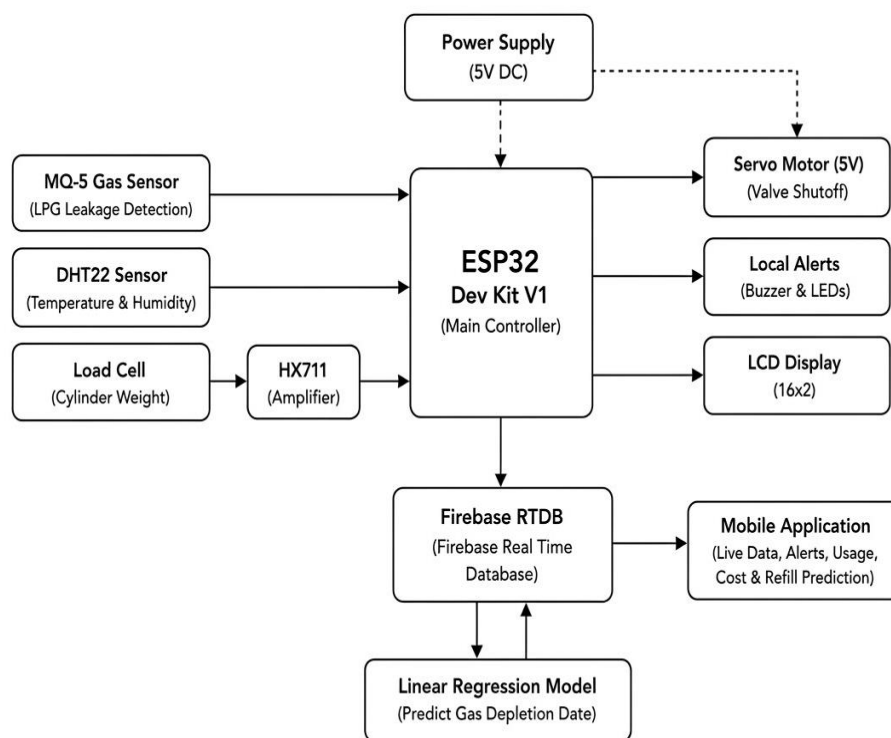


Figure 4.1: System Block Diagram

4.5 FLOWCHART OF SYSTEM OPERATION

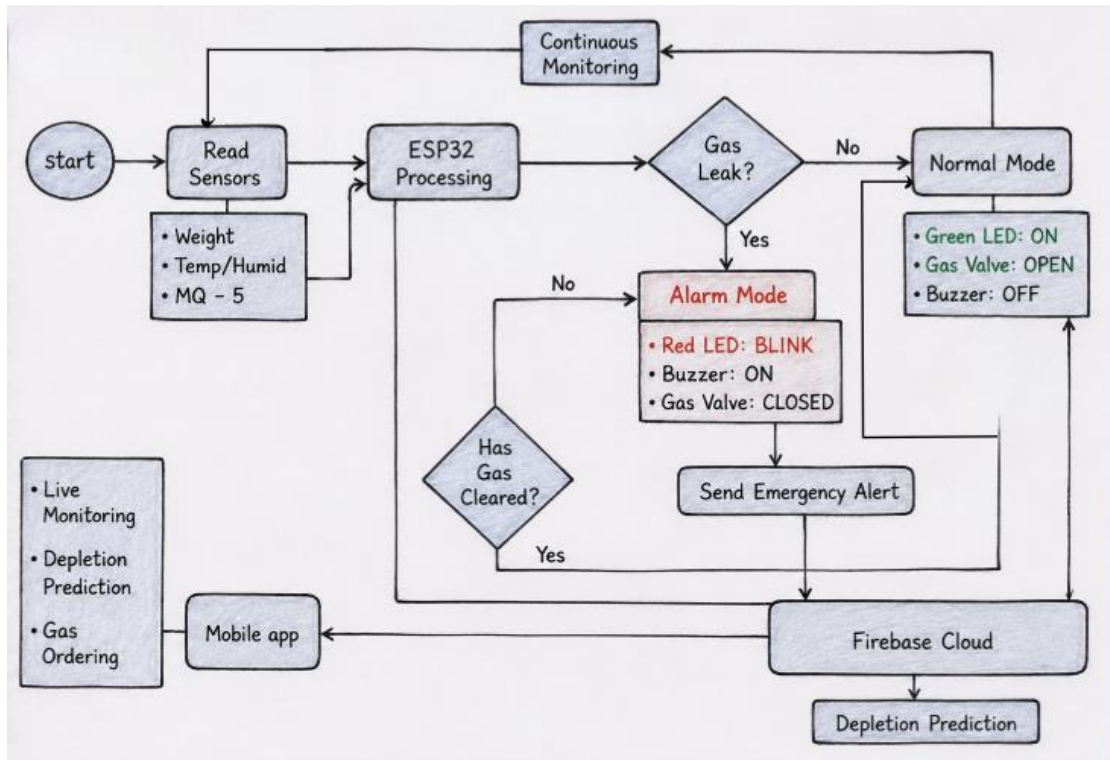


Figure 4.2 : Flowchart of the GasGuard System Operation

4.6 FLOWCHART OF MOBILE APPLICATION

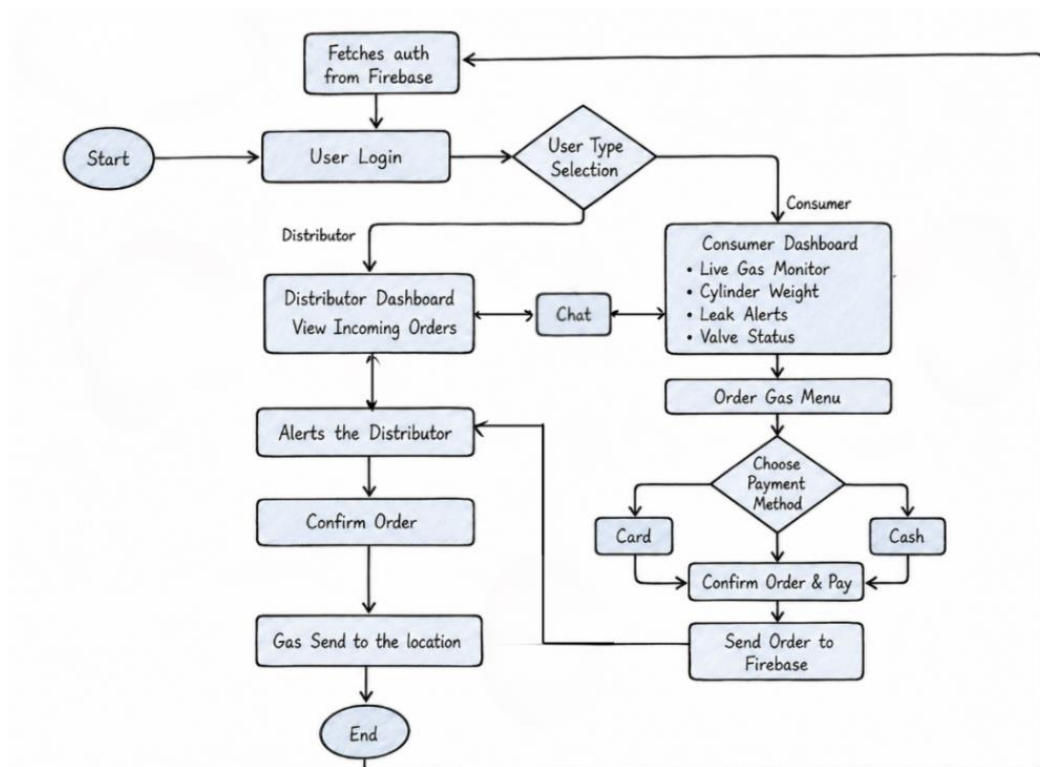


Figure 4.3 : Flowchart of the GasGuard Mobile Application

4.7 CALIBRATIONS

4.7.1 Load Cell Calibration

The load cell was calibrated using known weights to convert HX711 raw ADC readings into weight values. The platform was first tared to remove the zero-load offset, and then test weights were added while recording ADC readings. A linear calibration curve was obtained with a calibration factor of 92946.11 counts/kg, tare offset of 395001.12, and $R^2 = 0.9971$. This confirms that the load cell gives reliable readings for LPG cylinder weight monitoring.

The fitted calibration equation was:

$$y = 92946.11x + 395001.12$$

Where:

- y = Raw ADC count
- x = Added weight in kg
- **395001.12** = Tare offset
- **92946.11 counts/kg** = Calibration factor

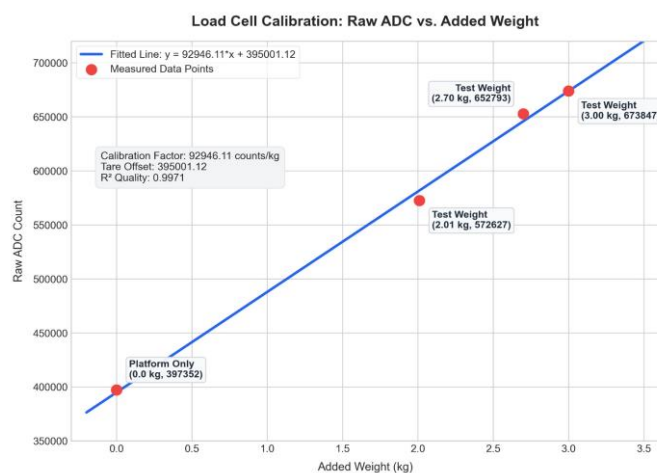


Figure 4.4 : Load Cell Calibration Graph

4.7.2 MQ - 5 Sensor Calibrations

The MQ-5 gas sensor was tested to find a safe gas leakage limit. First, the sensor was switched ON and kept in normal air for a few minutes to stabilize. Then the normal air reading was observed. After that, a small amount of LP Gas was brought near the sensor, and the sensor value increased.

Gas leak threshold was set to:

$$\text{Gas Threshold} = 500$$

Condition	MQ-5 Reading	System Status
Normal air	Below 500	Safe condition
Gas detected	Above 500	Leakage alert
Gas cleared	Below 500	Back to normal mode

Figure 4.1: MQ-5 Gas Leak Detection Calibration Results

4.8 CIRCUIT DESIGN

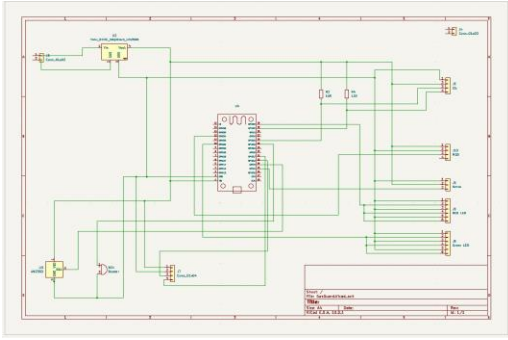


Figure 4.5: Schematic Diagram of PCB

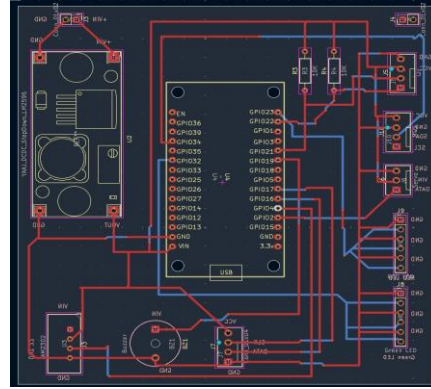


Figure 4.6: 3D View of PCB Layout View

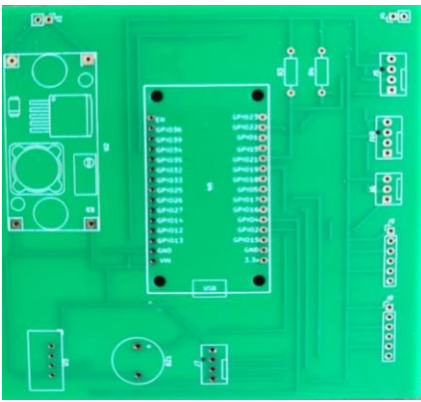


Figure 4.7: PCB

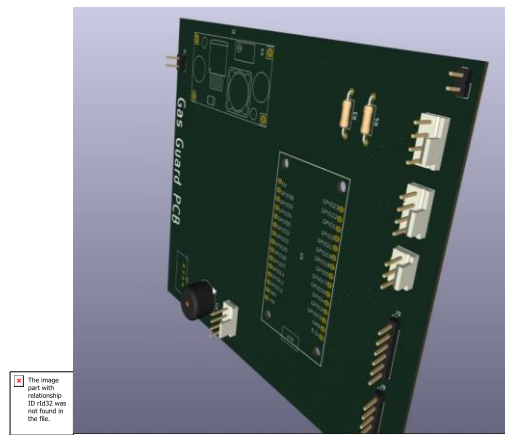


Figure 4.8: 3D View of PCB

4.9 GASGUARD MONITORING NODE DESIGN

The ESP32 DevKit V1 acts as the main controller of the GasGuard system, continuously reading and processing data from the load cell, HX711 module, MQ-5 gas sensor, and DHT22 sensor. The load cell and HX711 measure cylinder weight to estimate the remaining LPG quantity and monitor consumption patterns, while the DHT22 provides temperature and humidity data for environmental monitoring. These real-time sensor values are displayed locally on the 16×2 LCD.

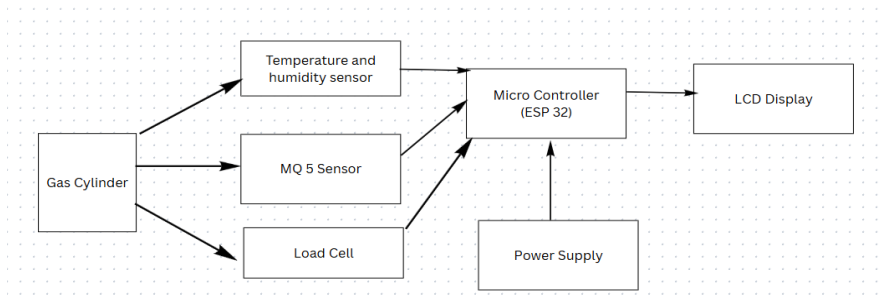


Figure 4.9: System Design of the GASGUARD Embedded Monitoring Node

4.10 GAS GUARD CONTROL UNIT DESIGN

The ESP32 DevKit V1 acts as the main controller of the control and actuation unit. It processes sensor readings and controls the LEDs, buzzer, LCD, and MG995 servo motor. When the MQ-5 detects LPG above the predefined safety threshold, the ESP32 activates local alerts and automatically closes the gas valve to reduce fire or explosion risk. Sensor and system status data are also sent to Firebase for remote monitoring through the mobile application.

The system is powered from a 230 V AC supply through an AC-DC adapter and LM2596 buck converter, which provides regulated DC power to the ESP32 Dev Kit V1 and other components for stable and safe operation.

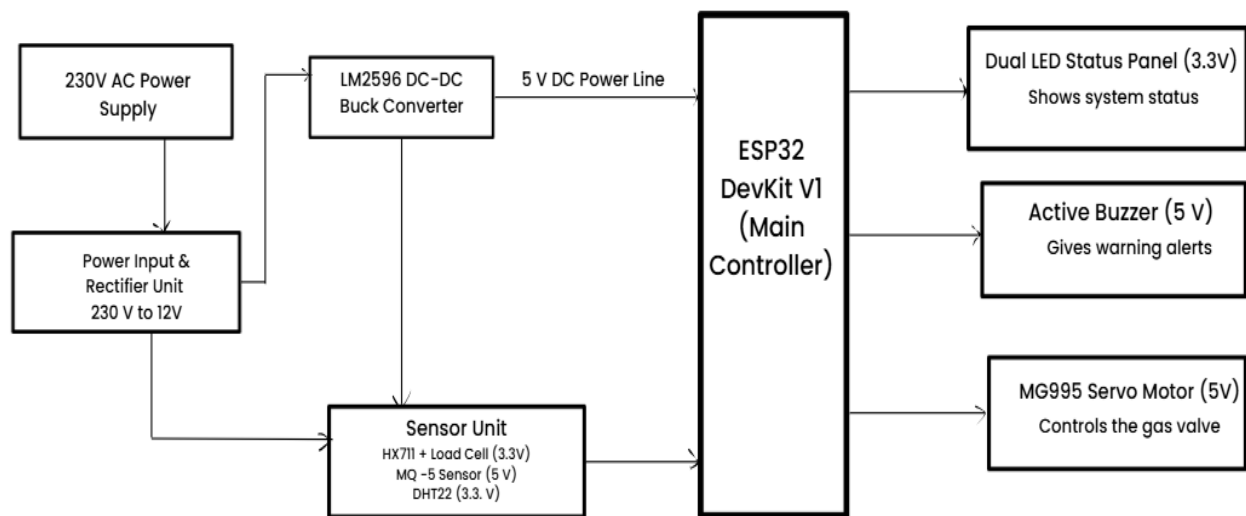


Figure 4.10: System Design of Control Unit

4.11 NETWORK COMMUNICATION PROTOCOLS AND CLOUD ARCHITECTURE & LOCATION SHARING

The GasGuard ESP32 operates in **Wi-Fi Station (STA) mode** and connects to a local internet gateway for communication with Firebase. **TCP** provides reliable data transmission, while **HTTPS** is used for secure authentication and Firebase database communication. The system periodically uploads sensor readings and system status to Firebase, while critical gas leak events are transmitted as safety records. In the current firmware, normal Firebase updates are performed every **5 seconds**.

Firebase Realtime Database acts as the central cloud platform for storing LPG level, gas leakage status, temperature, humidity, valve status, and other monitoring information. The stored data can then be accessed by the mobile application for live monitoring, historical analysis, and refill prediction.

Google Maps is integrated at the mobile-application level to support location-based LPG refill services. Users can select or share their delivery location when requesting a replacement cylinder, allowing registered suppliers or nearby LPG shops to identify the destination and coordinate delivery.

4.12 Communication System Design

The communication architecture of the GasGuard system operates in two main stages: device-to-cloud and cloud-to-user. The communication flow of GasGuard consists of two main stages: **device-to-cloud communication** and **cloud-to-user communication**. In the first stage, the ESP32 Micro Controller collects data from the MQ-5, DHT22, and load-cell processes the readings and sends updated information to Firebase through Wi-Fi. If a gas leakage condition is detected, the ESP32 performs the local safety response and records the event in Firebase.

In the second stage, the mobile application retrieves data from Firebase and presents LPG level, leakage status, temperature, humidity, usage information, cost estimates, safety alerts, and predicted refill dates. When a refill is required, the user can submit a request with a Google Maps-based delivery location, allowing the supplier to process and coordinate the cylinder delivery.

This integrated communication flow enables **real-time safety monitoring, remote access, predictive LPG management and refill coordination** within a single system.

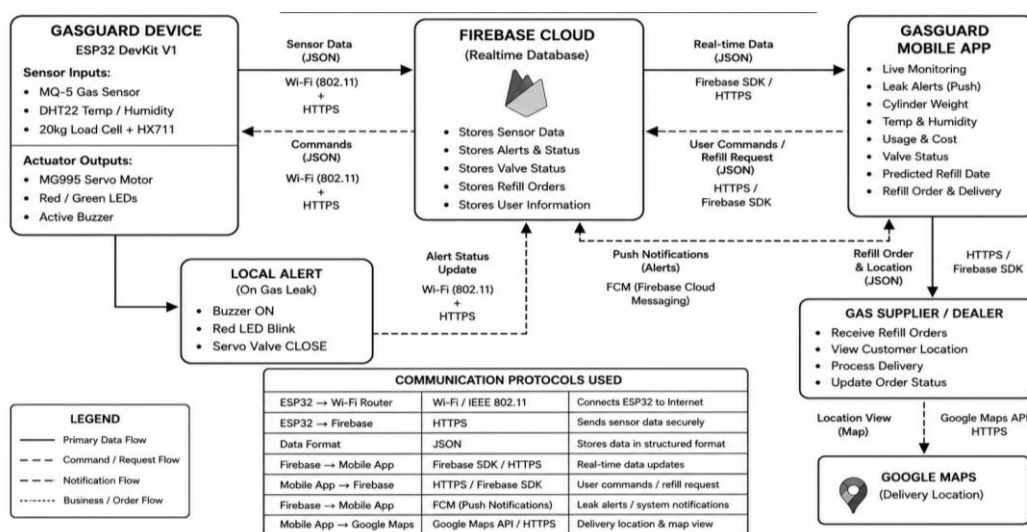


Figure 4.11: Communication System Design of the GASGUARD

4.13 Software Implementation

4.13.1 Embedded System Programming

The ESP32 DevKit V1 firmware was developed using the Arduino IDE and programmed to continuously acquire data from the load cell and MQ-5 sensor. The firmware processes sensor readings, compares them with predefined threshold values, and activates the safety response sequence when hazardous conditions are identified. This includes turning on the buzzer, lighting the warning LEDs, updating the LCD display, and driving the servo motor to shut off the gas valve.

4.13.2 Firebase Integration

Firebase is used as the cloud database for storing LPG level data, gas leakage alerts and historical usage information. The system sends sensor readings to Firebase in real time, enabling remote access through the mobile application. This cloud-based approach ensures that users can view updated system status, safety warnings and prediction results from anywhere with internet access.

4.13.3 Mobile Application Development

A cross-platform mobile application was developed using React Native to provide live monitoring and user-friendly visualization of the GasGuard system. The application displays LPG level updates, gas leak alerts, usage graphs, cost calculations, and predicted refill dates. It also includes Google Maps integration to allow users to share delivery locations and request refill services conveniently from the app.

4.14 Critical Design Decisions

Component selection	Chosen	Other options	Reason for selection
1) Gas Sensor	MQ-5 Gas Sensor	MQ - 2/ MQ - 6	1) Low cost 2) Simple to use 3) Easy ESP32 interfacing
2) Temperature & Humidity sensor	DHT22	DHT11 / LM35	1) Give Accurate temperature Value 2) Low cost
3) Main Controller	ESP 32 DevKit V1	Arduino Uno	1) Built-in Wi-F 2) processing capability 3) Multiple GPIO and ADC pins
4)Weight Measurement	20 kg Load Cell & HX711	FSR / Pressure Sensor	1) Accurate Weight Measurement 2) Low-cost digital interface
5).Valve Actuator	MG995 Servo Motor	MG90S/DC Motor	1) High torque 2) Precise angle control
6) Display	16×2 I ² C LCD	OLED Display	1) Low cost 2) Simple interface 3) Requires fewer GPIO pins

Table 4.2 : Critical Design Decisions

Firebase Realtime Database was selected as the cloud platform because it provides real-time data synchronization, cloud storage, and easy integration with the mobile application. Multiple Linear Regression was chosen for LPG depletion prediction due to its low computational complexity, simple implementation, and suitability for estimating remaining usage time. React Native was selected for mobile application development because it supports cross-platform development, Firebase integration and efficient user interface design.

4.15 CUSTOM WOODEN ENCLOSURE DESIGN

The GasGuard system was housed in a **custom-built wooden enclosure** to provide a low-cost, strong, and portable structure. The enclosure provides suitable space for the **ESP32, LCD display, status LEDs, buzzer, power circuitry, and internal wiring**. Its openable design also allows easy access for testing, maintenance, and future modifications.

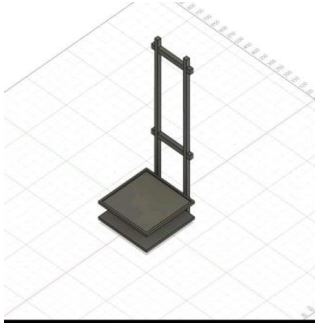


Figure 4.12 : Custom Wooden Enclosure Design

4.16 Prototype Development

The GasGuard prototype was assembled using commercially available components and tested in stages to verify the performance of each subsystem. First, the load cell and HX711 module were calibrated to ensure accurate cylinder weight readings. Then the MQ-5 sensor was tested for gas leakage detection, followed by verification of the servo motor response, buzzer alerts, and LCD output.

After individual testing, all components were integrated into the full prototype and connected to Firebase and the mobile application. The system was then evaluated for real-time monitoring, remote data logging, safety response, and Google Maps-based location support. This staged development approach helped identify and correct sensor calibration issues, power stability problems, and communication delays before final deployment.



Figure 4.13: Prototype Assembly Process

4.17 Final Implementation

The final GasGuard system consists of a fully integrated smart LPG monitoring unit built around the ESP32 DevKit V1, connected to the load cell, HX711, MQ-5 gas sensor, servo motor, buzzer, LEDs, and LCD display. The system operates continuously to monitor gas level, detect leakage, and activate automatic shutoff when necessary.



Figure 4.14: Final Product Design

4.18 MACHINE LEARNING BASED DEPLETION PREDICTION

4.18.1 Algorithm Selection and Theoretical Basis

This approach makes use of a supervised Machine Learning technique known as Multiple Linear Regression, which can be used to determine the time left before the exhaustion of LPG cylinders in days. Linear regression was chosen over neural network approaches owing to the linear decline in the amount of gas in an LPG cylinder during regular household usage.

4.18.2 Data Collection and Feature Engineering Pipeline

Real-time mass measurements are streamed from the HX711 load cell to Google Firebase via the ESP32 microcontroller. The model training pipeline utilizes a historical dataset of 180 records collected using a 5 kg LPG cylinder.

```

=====
FEATURE ENGINEERING & PREPROCESSING
=====

✓ Prepared 9 predictive features
- gas_weight_kg
- gas_percentage
- daily_usage_kg
- day_of_week
- weekend
- consumption_rate
- usage_velocity
- rolling_avg_7
- days_since_refill

✓ Feature matrix shape : (180, 9)
✓ Target vector shape  : (180,)
✓ Target                : days_remaining

```

Feature Engineering & Preprocessing

To capture usage trends, raw data was engineered into 9 predictive features (including **gas_weight_kg,gas_percentage,daily_usage_kg,day_of_week,weekend,consumption_rate,usage_velocity_rolling_avg_7,days_since_refill**). The dataset was divided into training (80%) and testing (20%) sets and normalized using StandardScaler to achieve zero mean and unit variance.

4.18.3 Model Training and Deployment Setup

Rather than an analytical single-step equation, the model was designed to be optimized iteratively using Gradient Descent over 100 epochs. Hyperparameter optimization via grid search established an optimal learning rate of 0.03. The regression logic executes server-side, fetching readings from Firebase and writing predictions back for display on the React Native mobile app .

4.19 COST ANALYSIS

Components	Cost
ESP32 Dev Kit V1	1050
Dual Channel HX711 Module	140
DHT sensor	360
20kg Load Cell	480
MQ-5 Gas Sensor	480
I ² C LCD Display 16×2	480
MG995 Servo Motor	1,150

12V DC Power Adaptor	600
Dot board (7cm x 5cm)	50
Buzzer/LED /Misc	300
PCB & Breadboard	3000
LM2596 3-40V to 1.5-35V 4A DC to DC Adjustable Step-Down Buck Module	200
Total	Rs. 8,290

Table 4.3: Component List and Cost Estimation

4.20 COST-SAVING MEASURES IMPLEMENTED

The project implemented several cost-effective strategies to reduce overall development expenses while maintaining system reliability and safety. Affordable, commercially available components such as the MQ-5 gas sensor (**LKR 480**), strain-gauge load cell (**LKR 480**), HX711 module (**LKR 140**) and LM2596 buck converter (**LKR 200**) were selected instead of expensive industrial monitoring equipment. A custom-built wooden cart enclosure was developed at a low material cost, providing a durable and portable housing solution. The use of a low-cost step-down power conversion system allowed the device to operate safely from a standard 230V AC supply without requiring expensive power management hardware.

The ESP32 was selected as the main controller instead of a Raspberry Pi due to its lower cost, built-in Wi-Fi, ADC support, and sufficient processing capability for embedded monitoring tasks. This eliminated the need for additional communication modules and reduced overall system complexity and cost.

The overall design focused on practical, reusable, and easily available components to ensure the system remained affordable for household and small commercial LPG applications.

4.21 ASSUMPTIONS AND LIMITATIONS

The system assumes stable Wi-Fi connectivity for Firebase synchronization, mobile monitoring, and remote notifications. Local safety functions, including leak detection, alarms, and automatic valve control, can operate without internet access.

Due to economic and availability constraints in Sri Lanka, obtaining a dedicated LPG cylinder for extended testing was difficult. Therefore, the load-cell system was initially tested using controlled weights up to approximately **5 kg**, representing different LPG gas quantities. These measurements were later treated as simulated gas-weight (using another weights) values for system calibration, monitoring, and prediction.

The machine learning model was trained using a **limited dataset**, so its predictions mainly represent the tested LPG consumption patterns. Prediction accuracy may vary with different cylinder capacities, household usage patterns, and larger real-world datasets.

MQ-5 readings may also vary with temperature and humidity, while load-cell measurements can be affected by vibration, uneven placement, and mechanical balance. Therefore, further testing with actual LPG cylinders, larger datasets, and different operating environments is required before commercial deployment.

4.21.1 Machine Learning Performance

Metric	Value
Prediction Purpose	LPG cylinder depletion time prediction
Model Type	Multiple Linear Regression
Input Features	9
Dataset Size	180 records
Training Configuration	100 epochs, learning rate 0.03
Testing R ² Score	98.83%
Prediction Error	MAE: 1.618 days, RMSE: 1.877 days

Table 4.4: Machine Learning Model Performance Summary

4.21.2 System Power Consumption

Component	Operating Current	Note
ESP32 DevKit V1	400 mA at 5V	Main controller with Wi-Fi active
MQ-5 Gas Sensor	170 mA at 5V	Gas sensor heater active
DHT22 Sensor	1.5 mA at 3.3V	Temperature and humidity sensing
HX711 + Load Cell	5 mA at 3.3V	Cylinder weight measurement
LCD + LED Indicators	70 mA at 3.3V	Display and status indication
Active Buzzer	60 mA at 3.3V	During gas leakage alert
MG995 Servo Motor	1,000 mA at 5V	During valve closing operation
Total Normal Operation	630 mA at 5V	Around 3.15 W continuous
Total During Alarm	1,700 mA at 5V	Around 8.5 W peak operation

Table 4.5: System Power Consumption Summary

4.21.3 End-to-End Alert Latency

Stage	Latency	Notes
MQ-5 gas sensor reading	~0.1 s	Analog value read from GPIO 34
Gas threshold detection	~0.1 s	Alarm triggers above 500 ADC value
LED/buzzer activation	0.2 s	Red LED and buzzer blink using 200 ms interval
LCD alert display	~0.2 s	Displays “ALERT : GAS LEAK”
Servo valve closing	1.5 s	MG995 rotates from 90° to 0°
Firebase Realtime Database event write	Network dependent	Depends on Wi-Fi speed
Mobile app alert synchronization	Network dependent	App receives Firebase update
Total local safety response	~1.7 s	LED/buzzer activation + servo valve closing
Total remote alert response	Network dependent	Depends on Wi-Fi, Firebase, and mobile app synchronization

Table 4.6: End-to-End Alert Latency Analysis

4.21.4 Overall System Performance

Metric	Value
Main Controller	ESP32 DevKit V1
Gas Detection Sensor	MQ-5 Gas Sensor
Gas Detection Threshold	500 ADC value
Sensor Reading Interval	1 second
Firebase Update Interval	5 second
Valve Open / Closed Angles	90° / 0°
Load Cell Calibration	92946.11 counts/kg, $R^2 = 0.9971$
Gas Capacity Used	5.0 kg
Prototype Cost	Rs. 8,290

Table 4.7: Overall System Performance Summary

CHAPTER 5: FINAL OUTCOME AND DISCUSSION

5.1 INTRODUCTION

This chapter describes the results of the initial outcome of the proposed GasGuard IoT and ML-Based Smart LPG Monitoring and Safety System. The prototype was tested in a controlled environment and in a simulated environment to assess accuracy of monitoring, safety response, monitoring of the cloud, the prediction and the mobile application capabilities. The results are presented on screenshots, tables, graphs and prototype pictures. Technical problems also discussed during the development and corrections taken.

5.2 FINAL OUTCOME

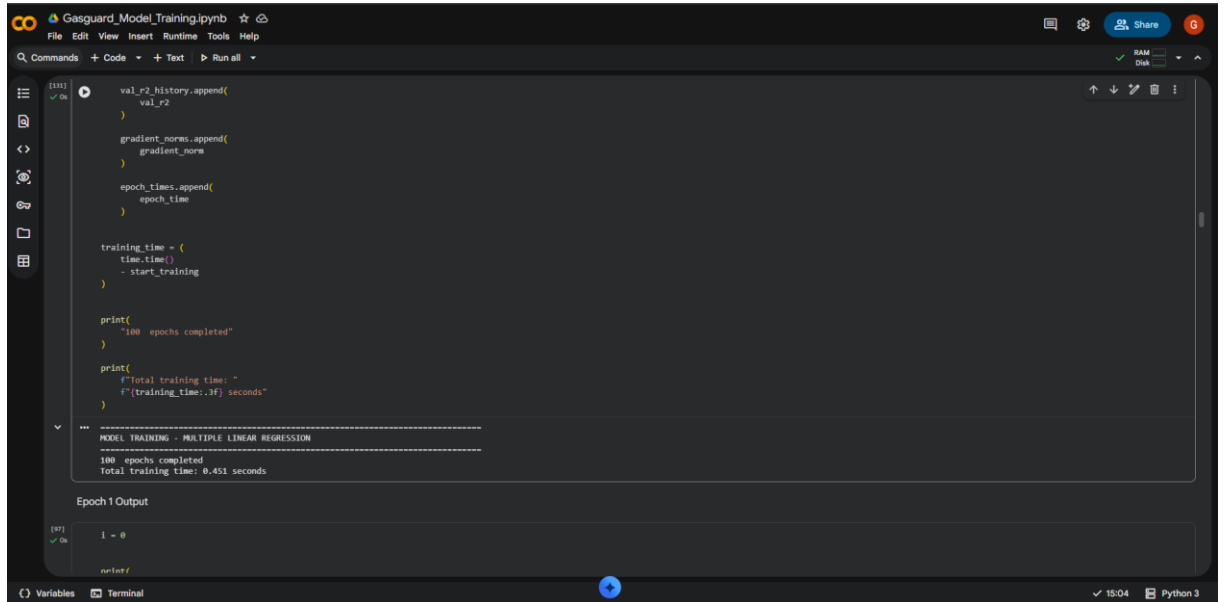
The project successfully developed a real-time IoT & ML -based LPG monitoring and safety system for household and small commercial use. The system provides continuous gas level monitoring, LPG leak detection, automatic valve control, Firebase-based cloud synchronization, predictive analysis, and mobile monitoring.

The final system includes:

- ESP32 DevKit V1 as the main controller for sensor processing and Wi-Fi communication.
- MQ-5 gas sensor for real-time LPG leakage detection.
- Load cell with HX711 amplifier for cylinder weight measurement and remaining gas estimation.
- DHT22 sensor for temperature and humidity monitoring.
- MG995 servo motor for automatic gas valve shutoff during leakage conditions.
- Buzzer, LEDs and 16×2 I²C LCD for local alerts and system status display.
- Firebase Realtime Database for real-time sensor storage, event logging, and remote synchronization.
- Linear Regression model for LPG depletion prediction and estimated refill dates.
- Mobile application with live monitoring, alerts, usage graphs, cost estimation, refill prediction, and LPG cylinder ordering.

5.3 SYSTEM OPERATION RESULTS

5.3.1 Model training Results



The screenshot shows a Jupyter Notebook interface with a dark theme. The notebook is titled 'Gasguard_Model_Training.ipynb'. The code cell contains the following Python code:

```
[111]: val_r2_history.append(
        val_r2
    )

    gradient_norms.append(
        gradient_norm
    )

    epoch_times.append(
        epoch_time
    )

    training_time = (
        time.time()
        - start_training
    )

    print(
        "100 epochs completed"
    )

    print(
        f"Total training time: "
        f"{training_time:.3f} seconds"
    )

    ...

MODEL TRAINING - MULTIPLE LINEAR REGRESSION
-----
100 epochs completed
Total training time: 0.451 seconds
```

The output cell shows the following text:

```
[111]: 1 = 0

print /
```

The bottom status bar indicates 'Variables', 'Terminal', '15:04', and 'Python 3'.

Figure 5.1: Machine Learning Model Training

5.3.2 Firebase Integration Results

Reading values from sensors, leakage information from the gas sensors, status of valves and events from the system have been uploaded successfully to Firebase Realtime database. The information has been synchronized with the React Native mobile app that enables the user to monitor the levels of LPG, temperature, humidity, and leakage status in almost real-time basis. Regular sensor readings have been uploaded every 5 seconds, whereas any leakage event has been uploaded immediately upon detection.

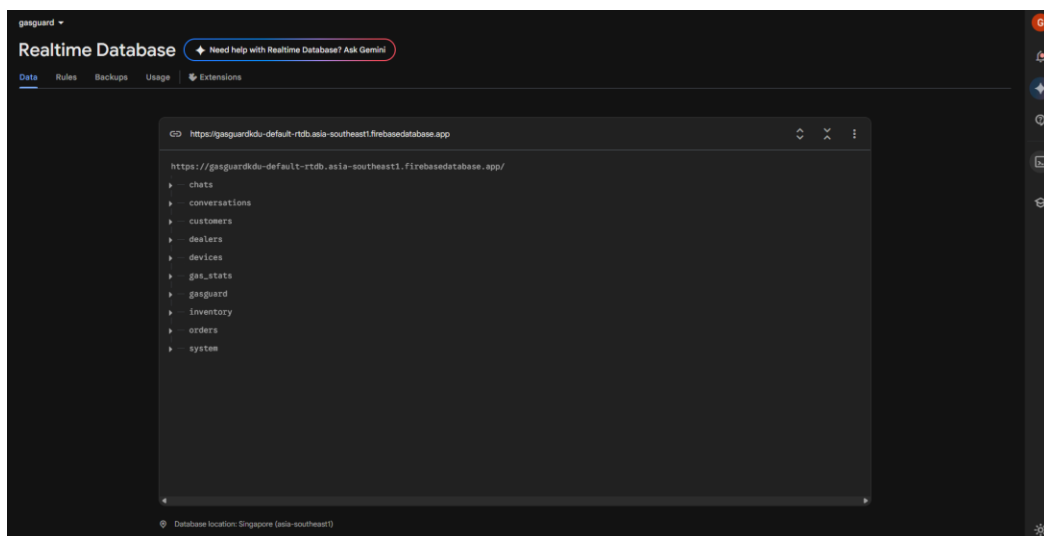
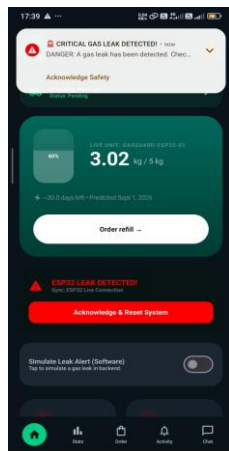


Figure 5.2: Firebase Integration Results

5.3.3 Mobile Application Results

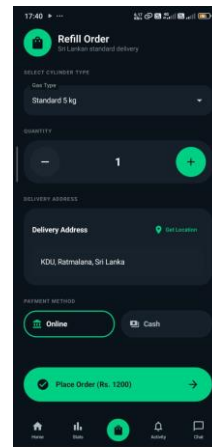
5.3.3.1 Customer Side Mobile Application



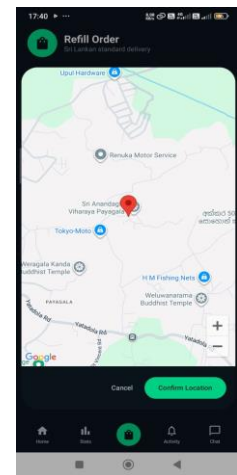
Gas Leak Alert



Home DashBoard



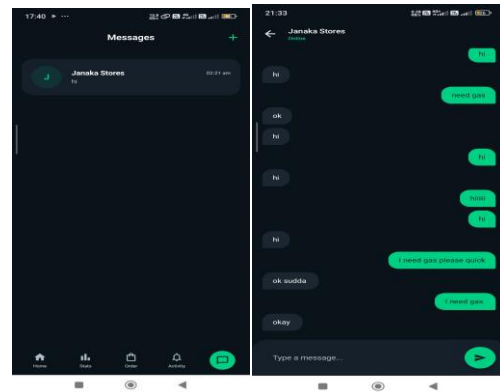
Refill Order



Delivery Location



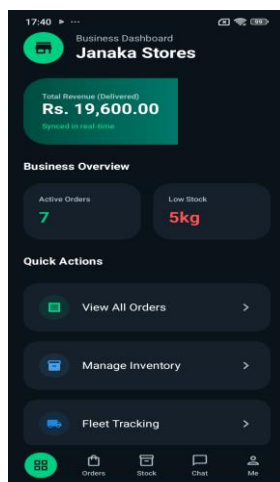
Usage Insights & Analytics



Real Time Chat

Figure 5.3: Customer Side Mobile App Dashboard

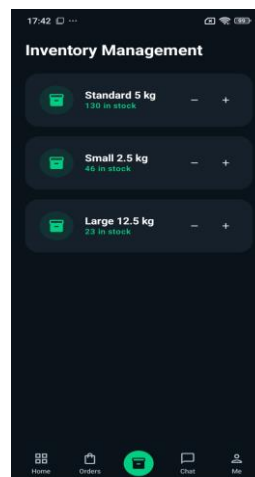
5.3.3.2 Dealer Side Mobile Application



Business DashBoard



Order Management



Inventory Management



Customer Chat

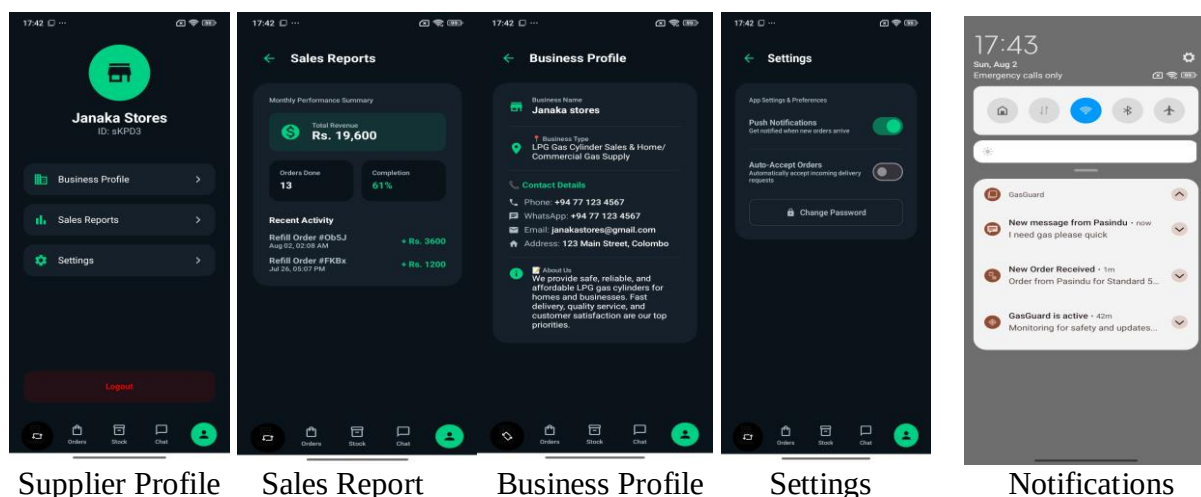


Figure 5.4: Dealer Side Mobile App Dashboard

The React Native mobile application featured a live display of LPG levels, gas leak detection, valve status, temperature, humidity, usage pattern, monthly expense, and Machine Learning depletion prediction. The app allowed for refill order placement, payment selection, location sharing through Google Maps, activity log and customer and supplier communication. Background notifications and location services offered gas leak warnings and delivery assistance even without opening the app.

Feature	Status	Notes
Real-time LPG monitoring dashboard	✓ Validated	LPG level, gas status, valve state, temperature and humidity
Gas leak alert system	✓ Validated	Gas leak warnings with foreground/background alert support
Usage analytics & cost estimation	✓ Validated	Consumption trends, gas history and monthly cost calculation
ML depletion prediction	✓ Validated	Estimated days remaining and predicted refill date
Refill ordering & location sharing	✓ Validated	Cylinder, payment method selection and Google Maps delivery location
Order/activity management	✓ Validated	Pending, accepted, delivered orders and activity history
Customer & supplier communication	✓ Validated	Real-time chat between users and registered LPG dealers
Supplier dashboard & inventory	✓ Validated	Revenue, active orders, stock levels and low-stock monitoring

Figure 5.1 : Functional Validation Summary of The Gas Guard Mobile APP

5.4 DISCUSSION

5.4.1 RESULTS

The system has been able to collect, monitor and show ambient gas concentration and cylinder weight data, which data is automatically wirelessly transmitted to the cloud back end using WiFi protocol. It also introduced the novel concept of effective automatic hazard mitigation and automatic utility tracking: it turned the gas shut-off valve off when it detected unsafe levels (alarm preparation), and it lit up various status LEDs and had a high torque mechanical servo motor that turns the valve off when an unsafe level was sensed (hazard detection).

The tracking interface remained very responsive and all depletion data, the costs computations, and notifications were accurate, shown on the local LCD display screen and remote mobile application Dashboard. In addition to this, the mobile application facilitates an easy ordering of a new LPG cylinder for delivery to your current place, once the gas level falls below a certain level to guarantee continuity in gas supply. Mobility of the mobile wooden cart turned out to be very good in terms of its portability and space utilization while meeting all other objectives like providing smartness to the communication system to improve security and monitoring control in the academic context of LPG was achieved.

5.4.2 DEPLETION PREDICTION MODEL RESULTS AND ANALYSIS

5.4.2.1 Training Convergence and Loss Analysis

The model demonstrated steady convergence during training, systematically minimizing the Mean Squared Error (MSE) loss across 100 epochs:

```
... [Epoch 1/100] Training Linear Regression model...  
    Forward pass completed  
    Computing gradients...  
    Updating weights with learning rate: 0.03  
    Current loss (MSE): 1035.5605  
    Validation score: -2.1597  
    Epoch completed in 0.0045s
```

Figure 5.5 : Training Convergence Log

Epoch	Training MSE	Loss Reduction	Validation R ²	Gradient Norm
1	1035.5605	Initial	0.3055	80.215120
25	70.3088	93.21%	0.9932	14.021500
50	27.7056	97.32%	0.9932	3.244684
75	25.0072	97.59%	0.9932	1.050468
100	24.5883	97.63%	0.9932	0.546706

```
[Epoch 25/100] Continuing training...
Convergence rate: 82.5%
Gradient norm: 14.021500
Loss reduced to: 70.3088 (-93.2%)
Validation score: 0.8275
Epoch completed in 0.0085s
```

Epoch 1 Training

```
[Epoch 50/100] Optimizing weights...
Adjusting regression coefficients
Feature importance recalculating
Loss reduced to: 27.7056 (-97.3%)
Validation score: 0.9742
Gradient norm: 3.244684
Epoch completed in 0.0037s
```

Epoch 50 Training

```
[Epoch 100/100] Final training pass...
Converged at final iteration
Final Loss (MSE): 24.5883
Gradient norm: 0.546706
Validation score: 0.9862
Epoch completed in 0.0093s
```

Epoch 100 Training

5.4.2.2 Evaluation Metrics and Cross-Validation

The trained model achieved high precision on both training and test data:

```
=====
MODEL EVALUATION & METRICS
=====

📊 Performance Metrics:

├─ R² Score (Training): 93.20%
├─ R² Score (Testing): 98.83%
├─ Mean Absolute Error: 1.618 days
├─ Root Mean Square Error: 1.877 days
├─ Mean Absolute Percentage Error: 12.99%
└─ Explained Variance: 98.83%

🔍 Cross-Validation Results (5-fold):

├─ Fold 1: 98.83%
├─ Fold 2: 98.85%
├─ Fold 3: 98.42%
├─ Fold 4: 98.41%
├─ Fold 5: 71.30%
└─ Mean CV Score: 93.16% (±10.93%)

🔺 Model Coefficients (Feature Importance):

├─ gas_weight_kg      : +7.9743 (POSITIVE impact on days remaining)
├─ gas_percentage     : +7.9743 (POSITIVE impact on days remaining)
├─ daily_usage_kg      : -0.2426 (NEGATIVE impact on days remaining)
├─ day_of_week         : +0.4340 (POSITIVE impact on days remaining)
├─ weekend              : +0.2702 (POSITIVE impact on days remaining)
├─ consumption_rate    : -0.4453 (NEGATIVE impact on days remaining)
├─ usage_velocity       : -0.7478 (NEGATIVE impact on days remaining)
├─ rolling_avg_7        : -0.4159 (NEGATIVE impact on days remaining)
├─ days_since_refill    : -1.2574 (NEGATIVE impact on days remaining)
└─ Intercept: +27.3480
```

Figure 5.6 : Evaluation & Performance Metrics

Training R² Score: 93.20%

Testing R² Score: 98.83%

Mean Absolute Error (MAE): 1618 days

Root Mean Square Error (RMSE): 1877 days

Mean Absolute Percentage Error (MAPE): 12.99%

5-Fold Cross-Validation Score: 93.16% ± 10.93

5.4.2.3 Model Comparison and Feature Importance

The baseline Linear Regression model was evaluated against regularized and non-linear variants:

Model	R ² Score	MAE	RMSE	Improvement vs Linear	Recommendation
Linear Regression	98.5230%	1.8255 days	2.1117 days	Baseline	Selected
Ridge	98.5108%	1.8288 days	2.1204 days	-0.0122%	Evaluated
Lasso	98.5338%	1.8199 days	2.1040 days	+0.0108%	Evaluated
Polynomial Degree 2	98.2314%	1.8231 days	2.3107 days	-0.2916%	Rejected

Table 5.2: Model Comparison

Model Comparison & Final Equation

MODEL COMPARISON				
Linear Regression	R ² 98.98%	MAE 1.504	RMSE 1.752	
Ridge (L2)	R ² 98.96%	MAE 1.516	RMSE 1.771	
Lasso (L1)	R ² 98.98%	MAE 1.512	RMSE 1.752	
Polynomial (deg=2)	R ² 98.70%	MAE 1.613	RMSE 1.983	
✓ Linear Regression selected as final model				

Days Remaining=58.161807+5.872535(gas_weight)+0.293627(gas_percentage)-65.797958(daily_usage)+0.043896(day_of_week)+0.385313(weekend)+34.309218(consumption_rate)-644.587024(rolling_avg_7)

Figure 5.7 : Model Comparison & Final Equation

A Multiple Linear Regression model was implemented using nine engineered features, StandardScaler preprocessing, and 100 real Gradient Descent epochs. The model achieved a testing R² score of 98.83%, an MAE of 1.618 days, an RMSE of 1.877 days, and a 5-fold cross-validation score of 93.16% ± 10.93% for LPG depletion prediction.

5.5 PROBLEMS ENCOUNTERED AND SOLUTIONS

1) UNSTABLE LOAD CELL READINGS

During early prototyping, the load cell produced unstable measurements with noticeable data drift due to external vibrations and sensor sensitivity. This affected LPG consumption calculations and gas depletion predictions. The issue was resolved by performing proper calibration using reference weights and applying software-based filtering techniques to improve measurement accuracy and stability.

2) POWER SUPPLY VOLTAGE INSTABILITY

Voltage fluctuations affected system reliability and risked damaging electronic components. A regulated buck converter was introduced to provide a stable DC supply and protect the controller and sensors.

3) I²C LCD COMMUNICATION AND CONFIGURATION ISSUES

During testing, the 16×2 I²C LCD showed black blocks or no text due to an incorrect address, poor soldering, and communication errors. The correct address was found using an I²C scanner, the clock was set to 100 kHz, and the contrast and connections were adjusted.

4) MQ-5 SENSOR ENVIRONMENTAL VARIATIONS

During testing, MQ-5 gas sensor readings varied with changes in temperature and humidity, affecting gas leak detection accuracy. The issue was resolved by testing the sensor under different environmental conditions and adjusting the detection threshold accordingly.

5) MG995 SERVO MOTOR POWER AND MOUNTING ISSUES

The MG995 servo motor caused voltage drops and power instability due to its high current demand during operation. Two servo motors were damaged during initial testing due to improper power supply. The problem was solved by using a regulated 5V power supply with an LM2596 module and providing a separate supply line with a common ground. A custom wooden servo mount was also developed to ensure proper alignment and reliable valve operation.

6) Limited Machine Learning Dataset

The depletion prediction model was initially trained using only 180 historical records collected from a standard 5 kg LPG cylinder. This limited the model's ability to generalize across different households, usage patterns, and cylinder sizes. The issue was reduced by applying feature engineering, train-test splitting, StandardScaler preprocessing, and 5-fold cross-validation. Future data collection will include additional cylinder types and longer usage periods.

5.6 TEST RESULTS SUMMARY

Test	Result	Notes
LPG leak detection	Pass	Safety response triggered when MQ-5 reading exceeded 500
Automatic valve shut-off	Pass	MG995 moved from 90° open to 0° closed , with 1.5 s movement allowance
Local alarm activation	Pass	Green LED remained ON during safe conditions, while the red LED and buzzer activated at 200 ms intervals while Leaks
Sensor monitoring	Pass	Weight, temperature and humidity readings updated every 1 s
Firebase data logging	Pass	Normal sensor/status data uploaded every 5 s when Wi-Fi was available
Load-cell calibration	Pass	HX711 tare performed using 10 readings for stable zero calibration

Table 5.3: Summary of Test Result

The GasGuard project managed to achieve its goals by successfully enabling the prediction of gas depletion using Machine Learning algorithms and gas control with the mobile application, while providing real-time LPG level monitoring, GasLeak detection and automatic valve control. If the MQ-5 reading is above 500, then the green LED will be off and red LED as well as buzzer will be on, with the MG995 Servo being turned on to move from 90° to 0° and close the gas valve on every 200ms. One second sensor readings were updated, and the normal Firebase data were synced every five seconds. This system would provide a simple and inexpensive solution for improving LPG safety, consumption monitoring and refill management in the domestic and small commercial sector.

CHAPTER 6: CONCLUSIONS AND FUTURE WORKS

6.1 CONCLUSION

The implementation of the “**GasGuard: IoT-ML Based Smart LPG Monitoring and Safety System**” provides a practical and cost-effective solution for improving LPG safety, consumption monitoring, and automated control in households [1], [2]. It is also suitable for small-scale commercial applications such as bakeries and food stalls in Sri Lanka.

The developed system integrates an ESP32 Dev Kit V1, MQ-5 gas sensor, and load cell to provide real-time LPG leak detection and cylinder level monitoring. Sensor data is transmitted through Wi-Fi to Google Firebase, enabling cloud monitoring and updates to the React Native mobile application. The system predicts gas depletion, sends safety alerts, and automatically activates the MG995 servo motor to shut off the gas supply during dangerous conditions.

The mobile application also allows users to order LPG refills, while nearby suppliers can receive and manage delivery requests. Overall, the system combines safety automation, predictive monitoring, and smart LPG management in a single platform.

6.2 PROJECT ACHIEVEMENTS

- Development of an ESP32 DevKit V1-based LPG monitoring system capable of real-time cylinder weight measurement, gas leakage detection, temperature and humidity monitoring.
- Integration of a Load cell with HX711 amplifier to estimate remaining LPG quantity and monitor gas consumption trends.
- Implementation of MQ-5 based LPG leak detection with automatic buzzer, LED, LCD warning and MG995 servo-controlled gas valve shutoff.
- Development of a complete Firebase Realtime Database pipeline for real-time sensor storage, event logging and remote synchronization.
- Development of a Mobile application providing live sensor data, LPG level monitoring, safety alerts, usage graphs, cost estimation and predicted refill dates.
- Implementation of a Linear Regression model achieving a testing **R² score of 98.83%**, an MAE of **1.618 days**, and a 5-fold cross-validation score of **93.16% ± 10.93%** for LPG depletion prediction.
- Implementation of a smart LPG refill management feature allowing households to request replacement cylinders and notify LPG suppliers when gas levels become low.
- Successful integration of DHT22 temperature and humidity data to support environmental monitoring and MQ-5 sensor evaluation.
- Development of a cost-effective prototype with an estimated total hardware cost of approximately **LKR 8,290**, making the system suitable for household and small commercial applications.

6.3 FUTURE WORKS

1. Integration of a Battery Backup System with Level Monitoring

The system can be improved by integrating a Lithium-Ion battery backup with charging management to maintain continuous operation during power failures. An ADC - based battery voltage reading circuit will also be included in this version, allowing the LCD interface and mobile dashboard to show the battery % in real time.

2. Multi-Sensor Expansion for Comprehensive Fault Detection

The system can be enhanced by integrating additional sensors, such as vibration and current sensors, to detect cylinder movement, external impacts, servo motor faults, and abnormal power consumption during operation.

3. Load Cell Data Filtering and Stability Improvements

The system may be improved by implementing a four-load-cell configuration instead of a single load cell to achieve balanced weight distribution and more accurate LPG level measurements. Advanced digital filtering techniques, such as moving average or Kalman filtering, can also be applied to reduce sensor noise caused by vibrations and improve measurement stability and precision.

4. Backup Communication During Internet Failure

The system can be enhanced by adding a GSM/LTE communication module to send critical gas-leak alerts when Wi-Fi or internet connectivity is unavailable. This would improve safety in areas with unstable network connections.

5. Smart Ventilation Control

An exhaust fan or ventilation system can be integrated to automatically activate when gas leakage is detected, helping to reduce gas accumulation in enclosed areas.

6.4 FINAL REMARKS

GasGuard Smart LPG Monitoring and Safety System is an LPG safety and usage management solution that integrates IoT, Firebase, Machine Learning, mobile application and automatic LPG valve control to enhance LPG safety and management. It offers a practical, low-cost and simple-to-use solution for households and small commercial users, allowing them to monitor the gas level, detect any leaks, track usage, predict gas refill, and receive alerts when it's running low. It is also capable of shutting off valves automatically when there are hazardous conditions of leakage for reducing the risk of fire and gas-related accidents. The system can be further enhanced for safer and more reliable LPG management with future improvements like automatic ventilation control.

CHAPTER 7: ANNEXURE

Annexure A: Source Code Repository (Full)

The complete firmware source code for the **GasGuard - IoT and ML-Based LPG Monitoring and Safety System** is maintained in a public GitHub repository. The repository includes the ESP32 firmware, sensor testing codes, React Native mobile application, machine learning model, Firebase database files, configuration files, and required libraries..

Repository URL:

<https://github.com/Lilendra-Pasindu-Harsha/GasGuard>

Repository Structure

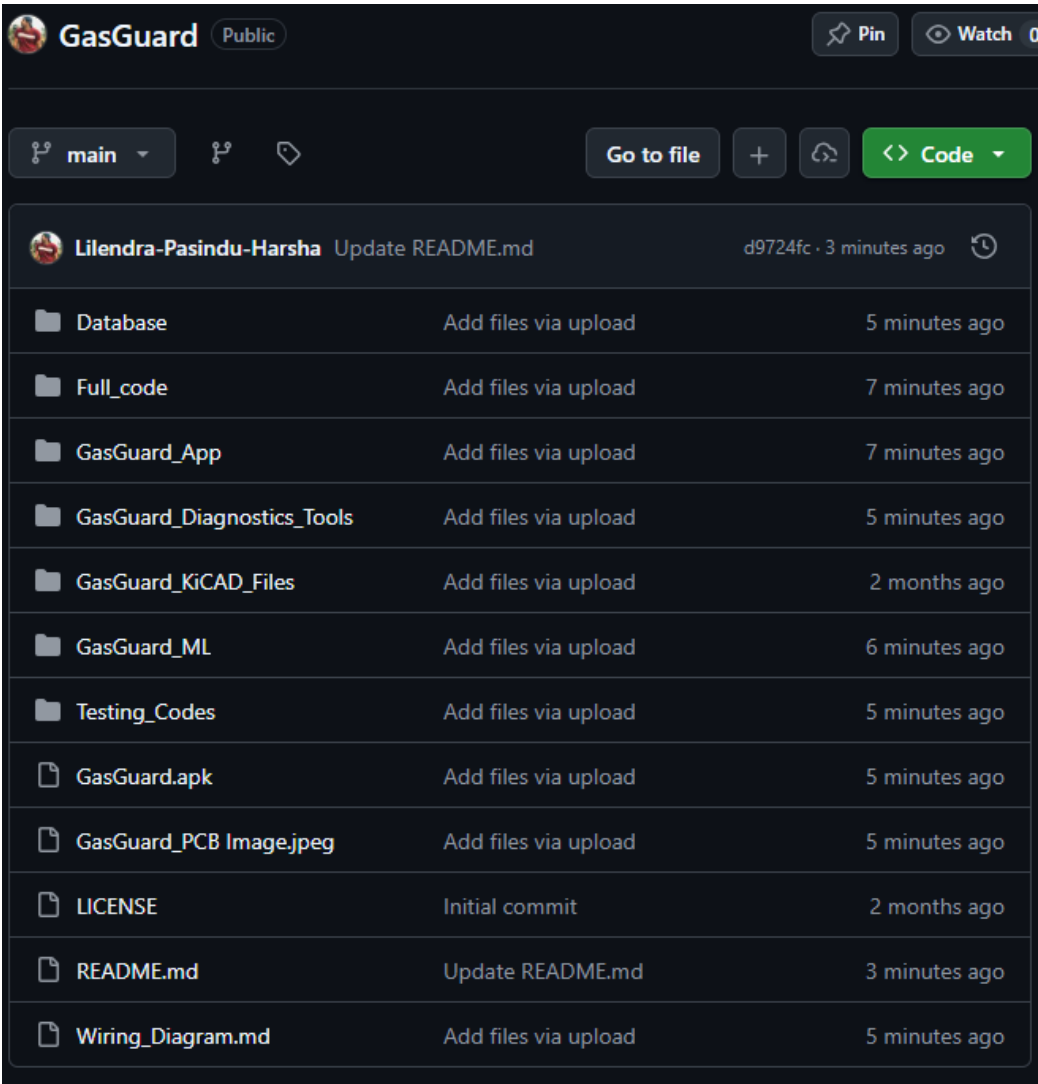
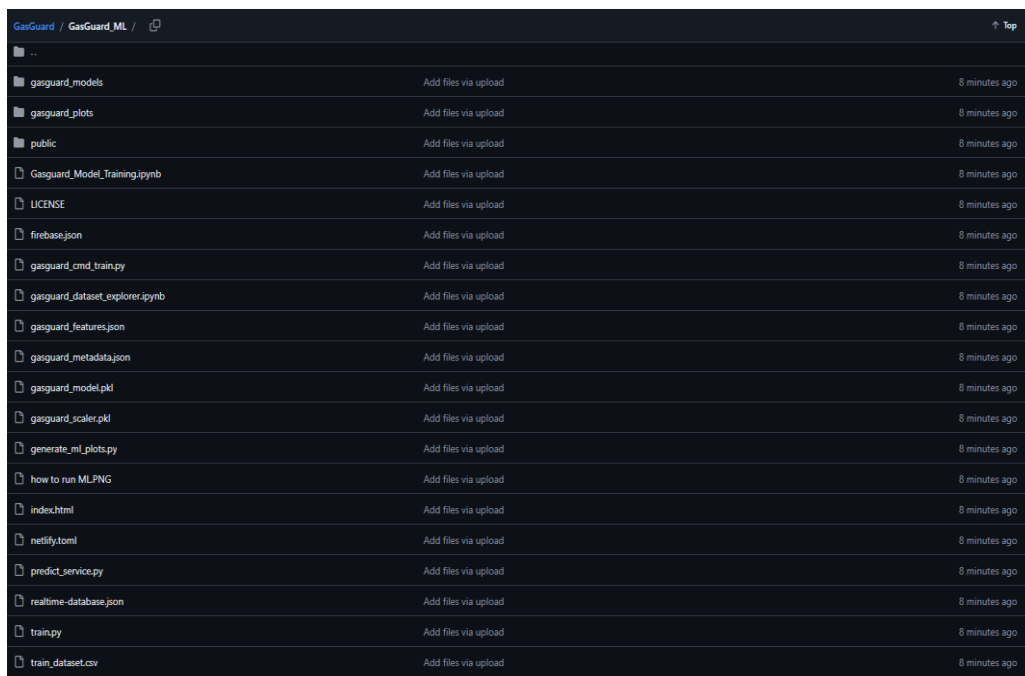


Figure Annexure A: Source Code Repository (Full)

Annexure B: GasGuard Machine Learning Model Strucuture.

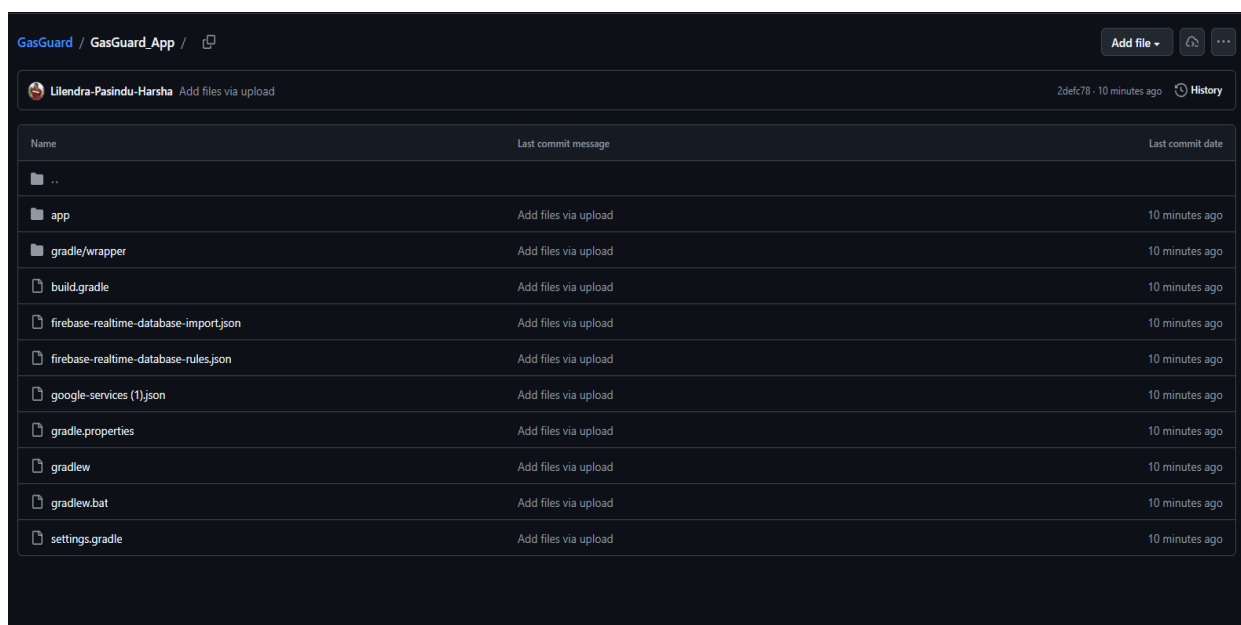


The screenshot shows a file explorer interface for the 'GasGuard / GasGuard_ML' repository. It lists various files and folders, each with an 'Add files via upload' button and a timestamp of '8 minutes ago'. The items include folders like 'gasguard_models', 'gasguard_plots', and 'public', and files like 'GasGuard_Model_Training.ipynb', 'LICENSE', 'firebase.json', 'gasguard_cmd_train.py', 'gasguard_dataset_explorer.ipynb', 'gasguard_features.json', 'gasguard_metadata.json', 'gasguard_model.pkl', 'gasguard_scaler.pkl', 'generate_ml_plots.py', 'how to run ML.PNG', 'index.html', 'netlify.toml', 'predict_service.py', 'realtime-database.json', 'train.py', and 'train_dataset.csv'.

Name	Last commit message	Last commit date
..		
gasguard_models	Add files via upload	8 minutes ago
gasguard_plots	Add files via upload	8 minutes ago
public	Add files via upload	8 minutes ago
GasGuard_Model_Training.ipynb	Add files via upload	8 minutes ago
LICENSE	Add files via upload	8 minutes ago
firebase.json	Add files via upload	8 minutes ago
gasguard_cmd_train.py	Add files via upload	8 minutes ago
gasguard_dataset_explorer.ipynb	Add files via upload	8 minutes ago
gasguard_features.json	Add files via upload	8 minutes ago
gasguard_metadata.json	Add files via upload	8 minutes ago
gasguard_model.pkl	Add files via upload	8 minutes ago
gasguard_scaler.pkl	Add files via upload	8 minutes ago
generate_ml_plots.py	Add files via upload	8 minutes ago
how to run ML.PNG	Add files via upload	8 minutes ago
index.html	Add files via upload	8 minutes ago
netlify.toml	Add files via upload	8 minutes ago
predict_service.py	Add files via upload	8 minutes ago
realtime-database.json	Add files via upload	8 minutes ago
train.py	Add files via upload	8 minutes ago
train_dataset.csv	Add files via upload	8 minutes ago

Figure Annexure B: GasGuard Machine Learning Model Strucuture

Strucuture Annexure C: GasGuard Mobile App Strucuture.



The screenshot shows a file explorer interface for the 'GasGuard / GasGuard_App' repository. It lists various files and folders, each with an 'Add files via upload' button and a timestamp of '10 minutes ago'. The items include folders like 'app' and 'gradle/wrapper', and files like 'build.gradle', 'firebase-realtime-database-import.json', 'firebase-realtime-database-rules.json', 'google-services (1).json', 'gradle.properties', 'gradlew', 'gradlew.bat', and 'settings.gradle'.

Name	Last commit message	Last commit date
..		
app	Add files via upload	10 minutes ago
gradle/wrapper	Add files via upload	10 minutes ago
build.gradle	Add files via upload	10 minutes ago
firebase-realtime-database-import.json	Add files via upload	10 minutes ago
firebase-realtime-database-rules.json	Add files via upload	10 minutes ago
google-services (1).json	Add files via upload	10 minutes ago
gradle.properties	Add files via upload	10 minutes ago
gradlew	Add files via upload	10 minutes ago
gradlew.bat	Add files via upload	10 minutes ago
settings.gradle	Add files via upload	10 minutes ago

Figure Annexure C: GasGuard Mobile App Strucuture

Annexure D : Library Dependencies

Library	Version	Purpose
LiquidCrystal_I2C	1.1.2	16×2 I2C LCD display control
DHT Sensor Library	1.4.6	DHT22 temperature and humidity sensor interface
HX711 Arduino Library	0.7.5	Load cell amplifier communication
ESP32Servo	3.0.5	MG995 servo motor PWM control
Firebase ESP Client	2.7.0	Firebase Realtime Database communication
WiFi	Built-in ESP32 Library	Wireless network connectivity
WIRE	Built-in Arduino Library	I2C communication protocol
Preferences	Built-in Arduino Library	Non-volatile storage for calibration values

Table 7.1: Library Dependencies

Annexure E : Component Datasheets

Component	Reference	PDF Link
ESP32 Dev Kit V1	AI-Thinker Technology Co., Ltd., <i>ESP32 Module Datasheet</i> , Version 1.3, 2018.	https://roboeq.ir/files/id/4034/name/ESP32%20MODULE.pdf
MQ-5 Gas Sensor	Winsen Electronics Technology Co., Ltd., <i>MQ-5 Semiconductor Sensor Manual</i> , Version 1.6, May 2021 .	https://www.alldatasheet.com/datasheet-pdf/pdf/1304543/WINSEN/MQ-5.html
HX711 Load Cell Amplifier	Avia Semiconductor, <i>HX711 24-Bit Analog-to-Digital Converter (ADC) for Weigh Scales Datasheet</i> , Version 2.0, 2018 .	https://www.alldatasheet.com/view.jsp?Searchword=HX711
20kg Load Cell	u-blox, "NEO-6 u-blox 6 GPS Modules Data Sheet," Doc No.	https://content.u-blox.com/sites/default/files/products/documents/NEO-6_DataSheet_(GPS.G6-HW-09005).pdf
DHT22 Sensor	Aosong Electronics Co., Ltd., <i>DHT22 (AM2302) Digital Temperature and Humidity Sensor Datasheet</i> , Version 1.3, 2018	https://cdn-shop.adafruit.com/datasheets/DHT22.pdf
MG995 Servo Motor	TowerPro, <i>MG995 High Torque Digital Servo Motor Datasheet</i> , 2019	https://www.towerpro.com.tw/product/mg995/

Table 7.1: Component Datasheet

REFERENCES

- [1] M. A. Baballe, U. Y. Magashi, B. I. Garko, A. A. Umar, Y. R. Magaji, and M. Surajo, "Automatic Gas Leakage Monitoring System Using MQ-5 Sensor," *Review of Computer Engineering Research*, vol. 8, no. 2, pp. 64–75, Dec. 2021, doi: 10.18488/journal.76.2021.82.64.75. Available: https://www.researchgate.net/publication/356812414_Automatic_Gas_Leakage_Monitoring_System_Using_MQ-5_Sensor
- [2] S. Kumar and R. Sharma, "IoT Based LPG Detection and Interfacing with ThingSpeak," *International Research Journal of Engineering and Technology (IRJET)*, vol. 7, no. 6, 2020. Available: https://www.academia.edu/44844118/IRJET_IOT_BASED_LPG_DETECTION_AND_INTERFACING_WITH_THINGSPEAK
- [3] E. F. Robalino P., A. Llerena, L. A. Flores, F. Trujillo, L. O. Freire, and F. Lara, "IoT-Based System for Detecting and Monitoring LPG Leaks in Residential Settings," *Engineering Proceedings*, vol. 115, no. 1, p. 9, 2025, doi: 10.3390/engproc2025115009. Available: https://www.researchgate.net/publication/397672259_IoT-Based_System_for_Detecting_and_Monitoring_LPG_Leaks_in_Residential_Settings
- [4] S. Kumar, G. Preet, N. Neetu, S. Shivani, M. Singh, and P. Priya, "IOT-Based Gas Leakage Detection and Gas Empty Alert System," *International Journal of Scientific Research in Engineering and Management (IJSREM)*, vol. 8, no. 6, pp. 1–9, Jun. 2024, Available: https://www.researchgate.net/publication/396898224_IOT-Based_Gas_Leakage_Detection_and_Gas_Empty_Alert_System
- [5] P. Priya, S. Kumar, G. P. Kaur, N. Neetu, et al., "IoT-Based Gas Leakage Detection and Gas Empty Alert System," *International Journal of Scientific Research in Engineering and Management (IJSREM)*, vol. 8, no. 6, pp. 1–9, Jun. 2024, doi: 10.55041/IJSREM35873. Available: https://www.researchgate.net/publication/396898224_IOT-Based_Gas_Leakage_Detection_and_Gas_Empty_Alert_System
- [6] "LPG Gas Leakage Detection Alert with Data Analysis Capability Using IoT Technology," 2024. Available: https://www.researchgate.net/publication/394580079_LPG_Gas_Leakage_Detection_Alert_with_Data_Analysis_Capability_Using_IOT_Technology